

Du Jianzhang

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GitHub <https://github.com/uu60> | CSDN blog <https://blog.csdn.net/Urakil?type=blog>

Java | C++ | Python | Concurrency | Microservices | Databases

Education Background

Indiana University 08/2024-Present

PhD in Computer Science

Advisor: Professor Chenghong Wang

Topic: Privacy database

The University of Hong Kong 09/2022-11/2023

Master of Science in Computer Science, GPA: 3.35 / 4.3

Northeastern University 09/2018-07/2022

Bachelor of Engineering in Software Engineering, GPA: 3.8 / 5.0 (87.507/100)

Research Experience

Parsec - 2PC Secure Database 08/2024-11/2025 (Under review)

Repo Link <https://github.com/uu60/parsec>

Contributions

- Designed and implemented a high-performance secure database system under the standard Two-Party Computation (2PC) model, enabling privacy-preserving query processing over distributed data without trusted third parties.
- Proposed a modular, layered architecture that decouples MPC primitives from relational database operators, improving extensibility and improved end-to-end system performance through parallel execution and secure network optimizations.
- Conducted in-depth research on the design and optimization of MPC and cryptographic protocols tailored to database workloads across diverse application scenarios.

Learned Index for Privacy Database 09/2024-10/2024 (On Arxiv)

Paper Link <https://arxiv.org/abs/2410.21164>

Contributions

- Designed and evaluated a learned index framework for privacy-preserving databases, leveraging machine learning models to accelerate query processing under differential privacy (DP) constraints.
- Optimized the generation of differentially private cumulative distribution functions (CDFs) for training data, enabling accurate and privacy-compliant model training.
- Implemented and compared multiple index models for efficient indexing, including piecewise linear regression and a dual-layer neural network.

Internship Experience

Trip.com (Shanghai) *Assistant of Back-end Development, Hotel R&D Department* 06-09/2023

Contributed to back-end development in the Merchant & Direct Connection Team, integrating data from hotel groups and suppliers, maintaining tax and rate-plan services, and improving system performance through Redis caching, multithreaded query processing, and migrating analytical workloads to ClickHouse for faster queries across the full software lifecycle.

Xiaomi Corporation (Beijing) *Software R&D, System Security Department* 05-08/2021

Contributed to Android development for the MIUI Security Center product, working on development for harassment interception and automatic task modules. Resolved multiple UI and back-end issues and led the end-to-end design and implementation of a new automatic task feature, addressing complex challenges such as circular task triggering and concurrency issues in asynchronous execution.

Honors and Awards

- Excellence Award in the 11th Blue Bridge Cup National Software and Information Technology Professional Talent Competition National Final (Java Software Development Undergraduate Group A), 11/2020
- First Prize in the 11th Blue Bridge Cup National Software and Information Technology Professional Talent Competition Liaoning Division (Java Software Development Undergraduate Group A), 10/2020
- National Scholarship for Academic Year 2018-2019, 12/2019
- Northeastern University First-class Scholarship and Outstanding Student Model, 12/2019